

Excel Formulas Reference Sheet

Arithmetic Formulas

- **Addition:** `=A1 + B1`
- **Subtraction:** `=A1 - B1`
- **Multiplication:** `=A1 * B1`
- **Division:** `=A1 / B1`
- **ROUND:** `=ROUND(A1, 2)` (Rounds the value in cell A1 to 2 decimal places)

Common Functions

- **SUM:** `=SUM(A1:A5)` (Adds the values in cells A1 to A5)
- **AVERAGE:** `=AVERAGE(A1:A5)` (Calculates the average of values in cells A1 to A5)
- **COUNT:** `=COUNT(A1:A5)` (Counts the number of cells with values in the range A1 to A5)
- **MAX:** `=MAX(A1:A5)` (Returns the maximum value in the range A1 to A5)
- **MIN:** `=MIN(A1:A5)` (Returns the minimum value in the range A1 to A5)

Text Functions

- **CONCATENATE:** `=CONCATENATE(A1, " ", B1)` (Joins the text in cells A1 and B1 with a space in between)
- **UPPER:** `=UPPER(A1)` (Converts the text in cell A1 to uppercase)
- **LOWER:** `=LOWER(A1)` (Converts the text in cell A1 to lowercase)
- **PROPER:** `=PROPER(A1)` (Capitalizes the first letter of each word in the text in cell A1)
- **TRIM:** `=TRIM(A1)` (Removes extra spaces from the text in cell A1)
- **SUBSTITUTE:** `=SUBSTITUTE(A1, "old", "new")` (Replaces occurrences of "old" with "new" in the text in cell A1)
- **LEN:** `=LEN(A1)` (Returns the number of characters in the text in cell A1)
- **LEFT:** `=LEFT(A1, 3)` (Returns the leftmost 3 characters from the text in cell A1)

- **RIGHT:** =RIGHT(A1, 3) (Returns the rightmost 3 characters from the text in cell A1)
- **MID:** =MID(A1, 2, 5) (Returns a substring from the text in cell A1, starting from the 2nd character and extracting 5 characters)

Logical Formulas

- **IF:** =IF(A1 > B1, "Yes", "No") (Checks if A1 is greater than B1 and returns "Yes" if true, "No" if false)
- **AND:** =AND(A1 > 5, B1 < 10) (Checks if both A1 is greater than 5 and B1 is less than 10)
- **OR:** =OR(A1 > 5, B1 < 10) (Checks if either A1 is greater than 5 or B1 is less than 10)
- **NOT:** =NOT(A1 > 5) (Checks if A1 is not greater than 5)
- **COUNTIF:** =COUNTIF(A1:A5, ">10") (Counts the number of cells in the range A1 to A5 that are greater than 10)
- **SUMIF:** =SUMIF(A1:A5, ">10", B1:B5) (Adds the corresponding values in the range B1 to B5 for cells in the range A1 to A5 that are greater than 10)
- **AVERAGEIF:** =AVERAGEIF(A1:A5, ">10", B1:B5) (Calculates the average of values in the range B1 to B5 if the corresponding value in the range A1 to A5 is greater than 10)
- **IFBLANK:** =IF(A1="", "Blank", "Not Blank") (Checks if cell A1 is blank and returns "Blank" if true, "Not Blank" if false)

Date and Time Functions

- **TODAY:** =TODAY() (Returns the current date)
- **NOW:** =NOW() (Returns the current date and time)
- **DATEDIF:** =DATEDIF(A1, B1, "d") (Calculates the number of days between dates in cells A1 and B1)

Lookup and Reference Functions

- **VLOOKUP:** =VLOOKUP(A1, A2:B10, 2, FALSE) (Searches for a value in column A and returns the corresponding value from column B)
- **HLOOKUP:** =HLOOKUP(A1, A2:F5, 3, FALSE) (Searches for a value in row 1 and returns the corresponding value from row 3)

- **INDEX:** =INDEX(A1:A10, 3) (Returns the value at the third position in the range A1 to A10)
- **MATCH:** =MATCH(A1, A1:A10, 0) (Finds the position of a value in the range A1 to A10)

Sort Data

1. **Filter:** Use the filter function to display only specific data based on criteria.
 - Select the range of data you want to filter.
 - Go to the Data tab and click on the Filter button.
 - Click on the filter dropdown arrows to select and display specific data.
2. **Sort:** Arrange data in a specific order based on one or more columns.
 - Select the range of data you want to sort.
 - Go to the Data tab and click on the Sort button.
 - Choose the column(s) you want to sort by and specify the sorting order (ascending or descending).
3. **Custom Sort:** Sort data based on custom criteria.
 - Select the range of data you want to sort.
 - Go to the Data tab and click on the Sort button.
 - In the Sort dialog box, click on the **Add Level** button to add additional sorting criteria.
 - Specify the column(s) and sorting order for each level.

Miscellaneous Functions

1. **Text to Columns:** Divide data in a single column into separate columns based on a delimiter.
 - Select the range of cells containing the text you want to split.
 - Go to the Data tab and click on the Text to Columns button.
 - Choose the **Delimited** option and click Next.
 - Select the delimiter that separates the data (e.g., comma, space, tab).
 - Choose the destination where you want the split data to be placed (e.g., new columns or existing columns).
 - Click Finish to split the text into separate columns.
2. **Data Validation:** Set rules to validate and control data entry in specific cells.
 - Select the cell(s) you want to create a rule for.
 - Go to the Data tab and click on **Data Validation**.

- Set the validation criteria, such as whole numbers, decimal numbers, dates, etc.
3. **Transpose Data:** Quickly change the orientation of data using the paste option.
- Copy the data you want to transpose.
 - Right-click on the cell where you want to paste the transposed data.
 - Choose the **Transpose** option under **Paste Options**.
4. **Multi-Sheet Formatting:** To format multiple sheets simultaneously:
- Press **CTRL + Click** on each sheet's tab that you want to format together.
 - Make your formatting changes in one sheet, and they will be reflected in all the selected sheets.